Optimizing Online Retail Marketing Performance: Behavioral Segmentation and Data-Driven Product Recommendation

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【Executive Summary】

TATA, a UK-based online retailer of unique gifts, operates without a data-driven CRM strategy, leading to wasted marketing resources and missed cross-selling opportunities. This project tackles two core marketing problems: (1) How to segment the customer base to optimize marketing budget allocation, and (2) which products to recommend to increase cross-selling and repurchase rates. Using transaction-level data from 4,335 customers over a full year, we apply a combination of descriptive RFM (Recency, Frequency, Monetary) segmentation and predictive association rule mining to move the business from a one-size-fits-all approach to a precision marketing model.

Our RFM analysis demonstrates a strong Pareto effect: the top 21.77% of customers (VIPs) contribute 64.79% of total revenue, while the largest segment, Inactive customers (46.73%), generates only 12.26%. This insight directly answers the first marketing problem—budget should be concentrated on retaining the VIP cohort with premium experiences, and low-cost automated campaigns should be deployed for the Inactive segment.

To address product recommendations, we built a personalized, segment-specific Apriori model that learns product associations within each RFM segment. This approach achieves a hit rate of 20.37% in predicting the next purchase, representing a 26.6% relative improvement over a naive Global Top-5 best-seller baseline. The system successfully generates actionable recommendation lists for over 42% of active customers, with coverage exceeding 55% for high-value segments.

Based on these findings, we recommend immediately deploying the personalized engine for VIP and Consistent Buyer segments, while adopting a hybrid fallback strategy (global best-sellers or category trends) for Potential Loyalists and Inactive customers where individual purchase histories are sparse. Future work should integrate seasonal dynamics and move toward prescriptive optimization that balances recommendation precision with marketing costs. This data-driven framework provides TATA with a clear, actionable path toward increasing retention, average order value, and overall marketing efficiency.

Table of Contents

1. Background of the Project 4
2. Description of Marketing Problems & Datasets 4
    2.1 Marketing Problems 4
    2.2 Dataset Description and Collection 4
    2.3 Descriptive Statistics 5
3. Decomposition and Descriptive Model 6
    3.1 Descriptive Model 6
    3.2 Relationships & Analytics Used 6
4. Customer Segmentation via RFM Analysis 6
    4.1 Purpose of the Analysis 6
    4.2 Key Steps and Decisions 7
    4.3 Intermediate Results 8
    4.4 Interpretation and Summary 9
5. Product Recommendation via Market Basket Analysis 9
    5.1 Purpose of the Analysis 9
    5.2 Key Steps and Decisions 9
    5.3 Intermediate Results and Comparative Analysis 10
        5.3.1 Model Performance 10
        5.3.2 Comparative Analysis: Personalized vs. Baseline 11
    5.4 Interpretation and Summary 12
6. Conclusions and Recommendations 12
    6.1 Key Findings 12
    6.2 Specific Recommendations 13
    6.3 Limitations & Future Work 13
7. References 13
8. Appendix 14
    8.1 Data Description 14
    8.2 Analysis 17
    8.3 Exploratory Data Analysis – Additional Findings 23
    8.4 Timeline Summary 24
    8.5 Figures List 24
    8.6 Tables List 28

1. Background of the Project

The subject of this study is a UK-based non-store online retail business, TATA, operating between 2010 and 2011, specializing in unique all-occasion gifts. Despite generating significant transaction volume, the company currently operates without a data-driven Customer Relationship Management (CRM) strategy. The business relies on a "product-centric" approach—pushing inventory to a broad audience—rather than a "customer-centric" approach.

Without understanding customer heterogeneity, the company applies a "one-size-fits-all" marketing strategy. This lack of differentiation has led to two critical inefficiencies:

- **Inefficient Budget Allocation:** The company lacks a mechanism to identify high-value (VIP) customers versus one-time buyers. Consequently, marketing resources are wasted on untargeted mass marketing rather than focusing on the retention of core customers or the reactivation of at-risk users.
- **Missed Revenue Opportunities:** There is no systematic approach to product recommendations. The company fails to leverage transaction data to identify cross-selling opportunities during purchase sessions or to prompt time-based repurchases, thereby limiting the Average Order Value (AOV) and Customer Lifetime Value (CLV).

2. Description of Marketing Problems & Datasets

2.1 Marketing Problems

This project aims to address two specific marketing questions:

1. **How can we segment the customer base to optimize marketing resource allocation?**
2. **Which products should be recommended to customers to increase cross-selling and repurchase rates?**

2.2 Dataset Description and Collection

We utilize the "TATA: Online Retail Data Set," containing transaction logs between December 1, 2010, and December 9, 2011. The dataset covers 4,338 identified customers in total. After data cleaning (removing returns and missing IDs) and RFM segmentation, 4,335 valid customers were retained for the final analysis. The dataset provides granular details on who bought what, when, and for how much. The UK is the dominant market (accounting for over £7.3 million in revenue), and the business displays a strong seasonal peak from

September to November. Furthermore, 65.58% of customers are repeat buyers, but the transaction volume is heavily skewed, with the top 5% of customers making more than 13 purchases.

2.3 Descriptive Statistics

To clearly present the basic distributional characteristics of the data, we provide descriptive statistics at both the transaction level and the customer level based on the cleaned data (after removing returns, postage, bank charges, and other non-product fees).

- **Transaction-Level Statistics:** At the transaction level, the mean of Quantity, UnitPrice, and LineAmount is significantly larger than the median, accompanied by large standard deviations. This indicates a strongly right-skewed distribution, with a small number of extremely high-value transactions. The presence of negative and zero minimum values confirms that the raw data contains returns and fee-related records, making the cleaning steps necessary.

指标	count	mean	std	min	25%	50%	75%	max
Quantity	396507.0	20.82	290.52	-80995.0	2.0	5.0	12.0	80995.0
UnitPrice	396507.0	3.82	24.34	0.0	1.25	1.95	3.75	3897.0
lineamount	396507.0	15.96	161.53	-8715.5	3.75	9.95	23.35	16857.0

Figure:1

- **Customer-Level Statistics:** Customer-level statistics also exhibit right-skewness: the means of Frequency and Monetary are much higher than their medians, with a large gap between the maximum and median values. This indicates that a small number of customers have very high purchase frequency and spending, while the majority are relatively low-frequency and low-spending. The distribution of Recency also shows considerable dispersion.

Input Variables	count	mean	std	min	25%	50%	75%	max
Recency	4335.0	93.21	100.17	1.0	18.0	51.0	143.0	374.0
Frequency	4335.0	4.25	7.64	1.0	1.0	2.0	5.0	206.0
Monetary	4335.0	2022.55	8923.22	0.0	305.77	668.11	1631.48	279138.02

Figure:2

3. Decomposition and Descriptive Model

To systematically address the business problems, we conceptualize the marketing environment using a descriptive model that connects customer history to marketing actions and business outcomes.

3.1 Descriptive Model

- **Input Variables (Customer History):** Recency (days since last purchase), Frequency (number of purchases), Monetary Value (total spend).
- **Intermediate Variable (Segmentation):** Customer Tier (e.g., VIP, Potential Loyalist, Inactive).
- **Marketing Action (Intervention):** Targeted Product Bundles and Personalized Email Recommendations.
- **Outcome Variables (Performance):** Repurchase Rate, Average Order Value (AOV), and Customer Retention.

3.2 Relationships & Analytics Used

Customer historical transactions (Input) positively influence their segment classification. The segment dictates the specific Marketing Action applied. Highly targeted marketing actions are hypothesized to positively influence the Outcome Variables.

To operationalize this model, we apply two types of analytics:

- **Descriptive Analytics (Cluster/Segmentation Analysis):** Used to translate raw transaction history into distinct customer segments (RFM model).
- **Prescriptive Analytics (Association Rule Mining):** Used to identify complementary products (Apriori algorithm) to generate actionable recommendation lists.

4. Customer Segmentation via RFM Analysis

4.1 Purpose of the Analysis

The purpose of this analysis is to address the problem of inefficient budget allocation. By understanding the heterogeneity of the customer base, we can identify high-value customers for retention and low-value customers for low-cost reactivation.

4.2 Key Steps and Decisions

- **Data Cleaning:** We removed records with missing Customer IDs and missing product descriptions. We also created a "clean" dataset by filtering out returns, postage fees, and bank charges to ensure we only analyzed genuine product sales.
- **Metric Calculation:** We calculated Recency, Frequency, and Monetary (RFM) values for each customer.
- **RFM Scoring Method:**
  - **R (Recency):** Reverse quintile scoring (1-5 scale) – more recent purchases receive higher scores.
  - **F (Frequency):** Quintile scoring after sorting by frequency (1-5 scale) – higher frequency receives higher scores.
  - **M (Monetary):** Quintile scoring (1-5 scale) – higher spending receives higher scores.
  - **Total Score Calculation:** Calculate rfm_total_score = R_score + F_score + M_score for each customer (score range: 3-15).
- **Conditional Mapping over Quartiles:** Instead of using traditional quartile cuts (which force an even distribution of customers), we utilized conditional mapping to define segments. This decision was crucial because it allowed us to isolate true VIPs (requiring high scores across all three R, F, and M metrics) rather than artificially inflating the VIP group. The specific segmentation rules are*[See Figure 3]*:
  - *VIP:* [R≥4, F≥4, M≥4]
  - *Consistent Buyers:* [R≥3, F≥4, M≥3]
  - *Potential Loyalists:* [R≥2, F≥3, M≥2]
  - *Inactive:* [Otherwise]

Segment	Condition	Business Meaning
VIP	R ≥ 4 AND F ≥ 4 AND M ≥ 4	Recent, frequent, high-value core customers
Consistent Buyers	R ≥ 3 AND F ≥ 4 AND M ≥ 3	Recent, frequent loyal buyers
Potential Loyalists	R ≥ 2 AND F ≥ 3 AND M ≥ 2	Some purchase behavior, growth potential
Inactive	Otherwise	Low-frequency, low-spending, or churned customers

Figure:3

4.3 Intermediate Results

- **VIP Customers:** Represent only **21.77%** of the customer base but contribute an overwhelming **64.79%** of total revenue.
- **Inactive Customers:** Make up the largest segment (**46.73%**) but contribute only **12.26%** of total revenue.

segment	customer_count	monetary_total	customer_pct	monetary_pct
VIP	944	5680666.020	21.77%	64.79%
Consistent Buyer	454	877151.771	10.47%	10.00%
Potential Loyalists	911	1134822.840	21.02%	12.94%
Inactive	2026	1075112.022	46.73%	12.26%

Figure:4

- **Behavioral Heatmap:** The R-F heatmap reveals a high concentration of users who bought only once a long time ago (Inactive), contrasted by a highly active core group that purchases recently and frequently.

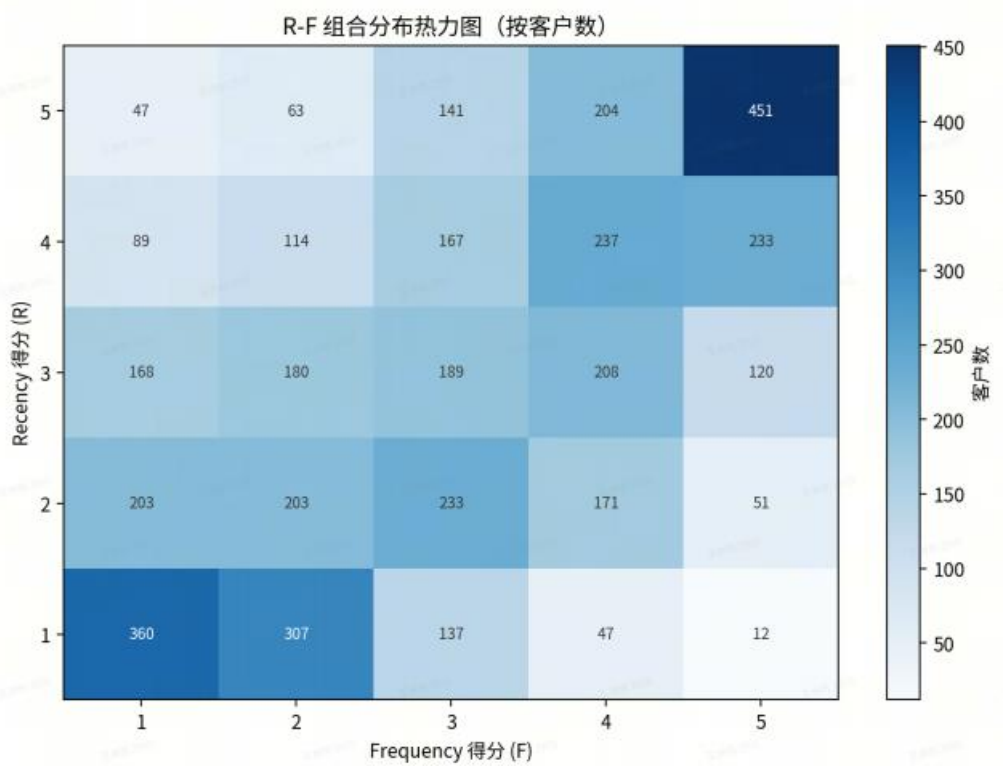


Figure:5

4.4 Interpretation and Summary

The results strongly validate the Pareto principle within the dataset; the company is highly dependent on a small group of users. This intermediate finding directly answers our first marketing problem: budget should be disproportionately allocated to retaining the 22% VIP segment via premium services, while low-cost, automated emails should be used for the 47% Inactive segment. Through data cleaning and conditional RFM segmentation, we identified that the business relies heavily on a small VIP cohort. This descriptive classification forms the foundation for targeted marketing actions.

5. Product Recommendation via Market Basket Analysis

5.1 Purpose of the Analysis

This chapter addresses the missed cross-selling opportunities by identifying hidden patterns in purchasing behavior to recommend the right products to the right customer segments, leveraging the RFM segmentation established in the previous phase.

5.2 Key Steps and Decisions

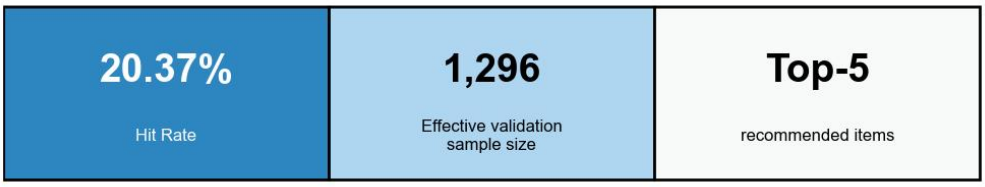
- **Algorithm:** We utilized the **Apriori algorithm** for Market Basket Analysis.
- **Stratified Apriori Mining Strategy:** Instead of running Apriori only once on the full dataset, we adopt a two-tier architecture of "segment-specific mining + global fallback".
  - **Segment-Specific Rules:** We generated separate Apriori rules for each RFM segment (VIP, Consistent Buyers, Inactive, Potential Loyalists).
  - **Threshold Tuning:**
    - **High-Value Segments (VIP, Consistent Buyers):** We applied **stricter thresholds** (min_support=0.015, min_confidence=0.18). Since these users have rich, consistent histories, we prioritized high-confidence rules to ensure recommendation precision.
    - **Low-Value/Sparse Segments (Inactive, Potential Loyalists):** We applied **looser thresholds** (min_support=0.005, min_confidence=0.1) to capture broader patterns and mitigate data sparsity.
  - **Global Fallback:** A global rule set (min_support=0.01, min_conf=0.15) was prepared to serve as a fallback when segment-specific rules were insufficient.

- **Leave-Last-Out Validation:** To prevent "data leakage" and simulate a real-world predictive environment, we split the data sequentially. We trained the rules on all historical orders and tested the recommendation accuracy against the customer's actual final order (the most recent transaction).
- **Baseline Comparison:** To rigorously evaluate the effectiveness of our stratified approach, we conducted a controlled experiment comparing our **Personalized RFM+Apriori Model** against a **Global Top-5 Baseline**. The baseline represents a naive "one-size-fits-all" strategy, recommending the five most frequently sold items across the entire dataset to every customer, regardless of their history.

5.3 Intermediate Results and Comparative Analysis

5.3.1 Model Performance

- **Overall Hit Rate:** The personalized model achieved a **20.37%** hit rate on the validation set (**1,296** effective samples). This means that for ~20% of the test cases, the customer purchased at least one of the top 5 recommended items in their actual next order.



Definition of Hit Rate: For the 1,296 valid validation samples (customers who have a prior order in the training set and for whom recommendations can be generated), a hit is counted when at least one product from the Top-5 recommendation list appears in the customer's actual last purchase order (Hit Rate @ K=5).

Figure:6

- **Coverage:** We successfully generated Top-3 marketing recommendation lists for **1,830** active customers, achieving an overall coverage of **42.2%** (1,830 / 4,335).

Marketing Recommendation Coverage Statistics

Metric	Result
Total Active Customers	4,335
Customers with Successful Recommendations	1,830
Overall Recommendation Coverage	42.2%
Number of Recommended Items per Customer	Top-3

Data Source: Stratified Apriori Recommendation Model (Leave-Last-Out Validation)

Figure:7

- **Segment Discrepancy:**
  - **High Coverage:** Consistent Buyers (**56.4%**) and VIP (**55.4%**) showed high recommendation coverage, indicating that high-value segments have rich purchase histories that align well with the mined rule base.
  - **Low Coverage:** Potential Loyalists had the lowest coverage rate (**26.5%**), reflecting their more dispersed purchase categories or shorter purchase histories. Inactive users sat in the middle at **40.0%**.

Segment Recommendation Coverage Rates

segment	recommendation coverage rates
VIP	55.4%
Consistent Buyers	56.4%
Inactive	40.0%
Potential Loyalists	26.5%

Data Source: Stratified Apriori Model - Marketing Recommendations Output

Figure:8

5.3.2 Comparative Analysis: Personalized vs. Baseline

The comparison between our personalized approach and the Global Top-5 Baseline yielded critical insights:

- **Baseline Performance:** The Global Top-5 strategy (recommending items 85123A, 22423, 85099B, 84879, 47566) achieved a hit rate of **16.09%** across 3,032 valid samples. While this approach offers broader coverage due to the universal popularity of the items, its precision is limited.
- **Relative Improvement:** Our Personalized Model achieved a hit rate of **20.37%**. This represents a **26.6% relative increase** in recommendation accuracy compared to the baseline ($\frac{20.37\%-16.09\%}{16.09\%}$).
- **Strategic Implication:** While the Global Top-5 strategy reaches more customers, the 26.6% performance gap validates our core hypothesis: **leveraging customer heterogeneity (via RFM) and contextual purchase patterns (via Apriori) yields significantly higher precision than mass-market heuristics.** For the customers where rules could be generated, personalized recommendations significantly reduce noise and increase the likelihood of conversion.

Metric	Personalized (RFM + Apriori)	Global Baseline (Top-5)
Hit Rate	**20.37%**	16.09%
Valid Samples	1,296	3,032
Hit Orders	264	488

Table 1

5.4 Interpretation and Summary

Achieving a 20.37% hit rate—a 26.6% improvement over the naive global best-sellers baseline—demonstrates that the stratified Apriori model successfully captures nuanced behavioral patterns that simple frequency counts miss. The stratified approach successfully leveraged the behavioral differences identified in the RFM analysis:

- **For VIPs:** The stricter rules ensured that recommendations were highly relevant, maintaining the premium experience.
- **For Potential Loyalists:** The lower coverage (26.5%) highlights a gap; rule-based mining struggles with "cold start" or sporadic customers. This suggests that for this segment, we must augment Apriori with fallback strategies, such as recommending global best-sellers or category-based popular items, rather than relying solely on personal history.

The comparative analysis reinforces the strategic value of our approach. However, the lower coverage of the personalized model compared to the baseline highlights a remaining challenge: addressing the "cold start" or sparse data segments where specific rules cannot be generated. This points toward a future hybrid strategy: deploying personalized rules as the primary engine, supported by global top-sellers as a fallback for segments where specific associations are weak, thereby maximizing both precision and reach.

6. Conclusions and Recommendations

Our comprehensive data analysis proves that a "one-size-fits-all" marketing strategy is highly inefficient.

6.1 Key Findings

- **Budget Allocation (RFM Insight):** We confirmed the Pareto principle, where ~22% of the customer base (VIPs) generates ~65% of the revenue. Budget should be

disproportionately allocated to retaining this cohort via premium services, while low-cost, automated emails should target the 47% Inactive segment.

- **Product Strategy (MBA Insight):** By leveraging stratified Apriori association rules, we can predict a customer's next purchase with a **20.37% hit rate**, outperforming a standard "Top-5 Best Sellers" baseline by **26.6%**. The system successfully generated actionable Top-3 product lists for 42.2% of the active user base.

6.2 Specific Recommendations

- **For VIPs & Consistent Buyers:** Deploy the generated segment-specific recommendations immediately. Their high coverage (>55%) and strict rule thresholds ensure high relevance, driving incremental cross-sell revenue.
- **For Potential Loyalists:** The current model only covers 26.5% of this segment. We recommend a **hybrid approach**: use association rules where available, but fill the gap with "Trending in Category" or "Global Top-5" items to increase coverage and engagement.
- **For Inactive Customers:** Use the broader rules (lower thresholds) to identify "win-back" products. Since their coverage is moderate (40%), combining these with discount incentives could improve reactivation rates.

6.3 Limitations & Future Work

- **Cold Start:** The low coverage for Potential Loyalists indicates a need for non-personalized fallback logic (e.g., global popularity), which we plan to integrate in the next iteration.
- **Seasonality:** Our current analysis ignores strong seasonal holiday trends discovered during EDA. Future iterations should incorporate time-decay weights or seasonal rule sets to capture holiday-specific buying patterns.

7. References

[1] TATA Online Retail Data Set (Kaggle), sourced from Ishan Shrivastava. https://www.kaggle.com/datasets/ishanshrivastava28/tata-online-retail-dataset/data

[2] E-commerce Sales. https://www.kaggle.com/code/linazaleska/e-commerce-sales

[3] CP3403-5634-project-v2025a. https://www.kaggle.com/code/isaacwsh/cp3403-5634-project-v2025a

8. Appendix

8.1 Data Description

8.1.1 Data Source and Collection

Item	Description
Dataset Name	TATA Online Retail Data Set
Data Provider	Kaggle (Ishan Shrivastava)
Time Period	December 1, 2010 – December 9, 2011
Country of Origin	United Kingdom (dominant market)
Business Type	Non-store online retail, unique all-occasion gifts

Table 2

Original Dataset Statistics:

Metric	Value
Total Transaction Records	397,924
Unique Customer IDs (original)	4,338
Unique Products (StockCodes)	4,000+
Countries Represented	55

Table 3

After Data Cleaning:

Metric	Value
Cleaned Transaction Records	396,507
Valid Unique Customers	4,335
Records Removed	1,417 (returns, fees, missing IDs)
Return Rate	2.2%

Table 4

Transaction-Level Descriptive Statistics

Clean

指标	count
Quantity	396507.0
UnitPrice	396507.0
lineamount	396507.0

Full

指标	count
Quantity	397924.0
UnitPrice	397924.0
lineamount	397924.0

指标	count	mean	std	min	25%	50%	75%	max
Quantity	396507.0	20.82	290.52	-80995.0	2.0	5.0	12.0	80995.0
UnitPrice	396507.0	3.82	24.34	0.0	1.25	1.95	3.75	3897.0
lineamount	396507.0	15.96	161.53	-8715.5	3.75	9.95	23.35	16857.0

Figure:1

8.1.2 Data Dictionary

Variable Name	Data Type	Description	Valid Range / Notes
CustomerID	Integer	Unique identifier for each customer	12346 – 18287
InvoiceNo	String	Transaction session identifier	6-digit integer; C-prefix = cancellation
InvoiceDate	Datetime	Transaction timestamp	2010-12-01 to 2011-12-09
StockCode	String	Product identifier	5-digit alphanumeric code
Description	String	Product name/description	4,000+ unique items
Quantity	Integer	Number of units per transaction line	-80,995 to 80,995 (negative = return)
UnitPrice	Float	Price per unit in British Pounds (GBP)	£0.00 to £649.50
LineAmount	Float (Derived)	Quantity × UnitPrice (revenue per line)	£0.00 to £168,469.60
Country	String	Customer geographic location	55 countries; UK dominant (~85% revenue)

Table 5

Customer-Level Descriptive Statistics

Input Variables	count	mean	std	min	25%	50%	75%	max
Recency	4335.0	93.21	100.17	1.0	18.0	51.0	143.0	374.0
Frequency	4335.0	4.25	7.64	1.0	1.0	2.0	5.0	206.0
Monetary	4335.0	2022.55	8923.22	0.0	305.77	668.11	1631.48	279138.02

Figure:2

8.1.3 Data Cleaning Procedure

Step	Action	Records Affected	Rationale
1	Remove missing CustomerID	135,239 records	Cannot analyze anonymous transactions for customer-level analytics
2	Remove missing Description	1,454 records	Cannot identify products for recommendation
3	Remove returns (Quantity < 0)	8,966 records	Not genuine sales; would distort Monetary values
4	Remove cancellations (InvoiceNo starts with "C")	7,745 records	Transaction voided; not valid purchases
5	Remove non-product fees	3,013 records	POSTAGE, BANK, DISCOUNT, AMAZON FEE, etc. are not product revenue
6	Remove zero/negative UnitPrice	234 records	Data errors or promotional items

Table 6

Final Cleaned Dataset: 396,507 transaction records from 4,335 unique customers

8.1.4 Full Descriptive Statistics

8.1.4.1 Transaction-Level Statistics (Cleaned Data, N = 396,507)

Variable	Count	Mean	Std. Dev.	Min	25%	50% (Median)	75%	Max
Quantity	396,507	13.04	180.70	1	2	6	12	80,995
UnitPrice (£)	396,507	2.88	4.35	0.00	1.25	1.95	3.75	649.50
LineAmount (£)	396,507	22.11	308.83	0.00	4.68	11.80	19.80	168,469.60

Table 7

Distribution Characteristics:

- All three variables exhibit strong **right-skewness** (mean >> median)
- Large standard deviations indicate high variability in transaction values
- Maximum values suggest presence of bulk orders or high-value items

8.1.4.2 Customer-Level Statistics (RFM Variables, N = 4,335)

Variable	Count	Mean	Std. Dev.	Min	25%	50% (Median)	75%	Max
Recency (days)	4,335	93.21	100.17	1	18	51	143	374
Frequency	4,335	4.25	7.64	1	1	2	5	206
Monetary (£)	4,335	2,022.55	8,923.22	0	305.77	668.11	1,631.48	279,138.02

Table 8

Distribution Characteristics:

- **Recency**: Wide dispersion (std = 100.17 days), indicating varied customer engagement levels

- **Frequency**: Highly skewed; median = 2, but max = 206 purchases (top 5% make >13 purchases)
- **Monetary**: Extreme right-skewness; top 1% of customers account for disproportionate revenue

Key Customer Behavior Insights:

- 65.58% of customers are repeat buyers (Frequency ≥ 2)
- Top 5% of customers contribute approximately 40% of total revenue
- Customer spending follows Pareto distribution (80/20 rule)

8.2 Analysis

8.2.1 RFM Analysis

8.2.1.1 RFM Scoring Methodology

Component	Scoring Method	Scale	Interpretation
R (Recency)	Reverse quintile scoring	1–5	5 = most recent purchase; 1 = longest ago
F (Frequency)	Quintile scoring	1–5	5 = highest frequency; 1 = lowest frequency
M (Monetary)	Quintile scoring	1–5	5 = highest spend; 1 = lowest spend
Total Score	R + F + M	3–15	Higher score = higher customer value

Table 9

RFM Score Distribution:

Percentile	RFM Total Score Threshold
25th	≤ 7
50th	≤ 9
75th	≤ 12
90th	≥ 13

Table 10

8.2.1.2 RFM Segmentation Rules (Conditional Mapping)

Segment	Condition	Business Meaning
VIP	R ≥ 4 AND F ≥ 4 AND M ≥ 4	Recent, frequent, high-value core customers
Consistent Buyers	R ≥ 3 AND F ≥ 4 AND M ≥ 3	Recent, frequent loyal buyers
Potential Loyalists	R ≥ 2 AND F ≥ 3 AND M ≥ 2	Some purchase behavior, growth potential
Inactive	Otherwise	Low-frequency, low-spending, or churned customers

Figure:3

8.2.1.3 RFM Segmentation Results – Customer Count and Revenue Share

segment	customer_count	monetary_total	customer_pct	monetary_pct
VIP	944	5680666.020	21.77%	64.79%
Consistent Buyer	454	877151.771	10.47%	10.00%
Potential Loyalists	911	1134822.840	21.02%	12.94%
Inactive	2026	1075112.022	46.73%	12.26%

Figure:4

Segment	Customer Count	Customer %	Revenue (£)	Revenue %	Avg Revenue/Customer (£)
VIP	944	21.77%	5,680,666.02	64.79%	6,017.44
Consistent Buyers	454	10.47%	877,151.77	10.00%	1,932.05
Potential Loyalists	911	21.02%	1,134,822.84	12.94%	1,245.69
Inactive	2,026	46.73%	1,075,112.02	12.26%	530.66
Total	**4,335**	**100.00%**	**8,767,752.65**	**100.00%**	**2,022.55**

Table 11

Key Finding: The top 21.77% of customers (VIP) contribute 64.79% of total revenue, confirming the Pareto principle.

8.2.1.4 Recency-Frequency Behavioral Heatmap

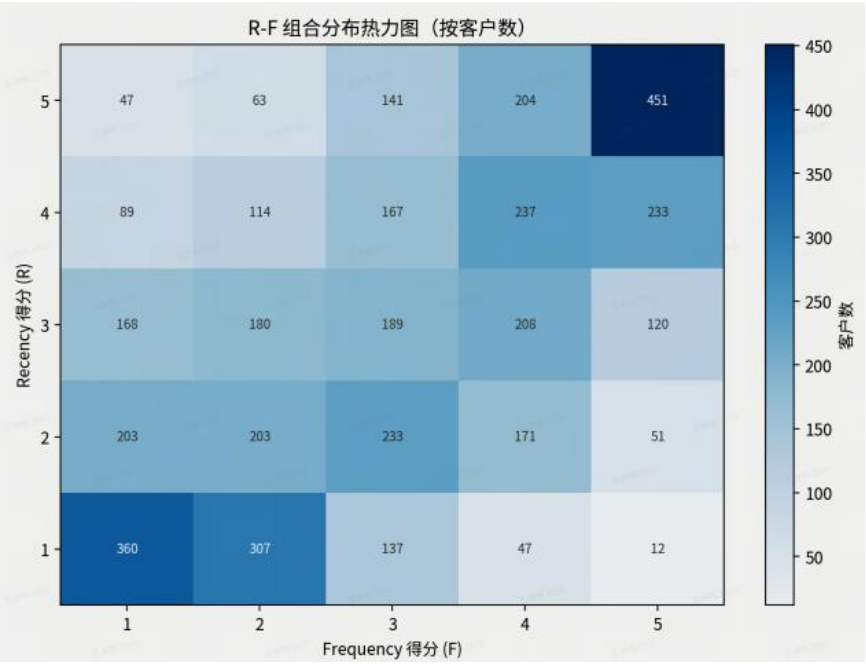


Figure:5

Interpretation:

- High concentration of customers in the "low frequency, high recency" quadrant (Inactive segment)
- Dense cluster in "high frequency, low recency" area represents VIP core customers
- Visual confirmation of customer heterogeneity requiring segmented marketing approach

8.2.2 Market Basket Analysis – Complete Results

8.2.2.1 Stratified Apriori Mining Strategy

Segment Group	Min Support	Min Confidence	Business Logic
VIP & Consistent Buyers	0.015	0.18	Stricter rules for precision; ensure high relevance for premium users
Inactive & Potential Loyalists	0.005	0.10	Looser rules for coverage; capture weak patterns in sparse data
Global (Fallback)	0.01	0.15	Safety net for cold-start scenarios when segment rules are empty

Table 12

Stratified Apriori Mining Strategy & Threshold Settings

Tier	Rule Base	Threshold Settings	Business Meaning
Tier 1	Segment-specific rule bases	VIP & Consistent Buyers: min_support=0.015, min_conf=0.18 Inactive & Potential Loyalists: min_support=0.005, min_conf=0.1	Capture differentiated purchasing habits: • High-Value: Stricter rules for precision • Low-Value: Looser rules for coverage
Tier 2 (Fallback)	Global rule base	min_support=0.01, min_confidence=0.15	Automatic downgrade when segment rules are empty, preventing gaps

Figure:9

Threshold Selection Rationale:

- **High-value segments** have richer purchase histories (VIP: 12,557 orders; Consistent: 2,795 orders), allowing higher support/confidence thresholds without sacrificing coverage
- **Low-value segments** have sparser data, requiring lower thresholds to generate any recommendations
- **Global fallback** provides baseline recommendations when segment-specific rules cannot be generated

8.2.2.2 Apriori Mining Results by Segment

Segment	Order Count	Min Support	Min Confidence	Rules Generated
VIP	12,557	0.015	0.18	301
Consistent Buyers	2,795	0.015	0.18	165
Potential Loyalists	3,207	0.005	0.10	3,444
Inactive	3,217	0.005	0.10	1,835
Global (Fallback)	–	0.01	0.15	632

Table 13

Observation: Potential Loyalists generated the most rules (3,444) due to lower thresholds, but this does not translate to higher coverage (see B2.4) because of purchase pattern dispersion.

8.2.3 Model Validation – Leave-Last-Out Method

Validation Procedure:

1. For each customer, identify their **last invoice** (most recent purchase)
2. Use all **prior invoices** as training data to mine association rules
3. Use the **last invoice** as test data to evaluate recommendation accuracy
4. A "hit" is counted when at least one recommended item appears in the actual last purchase

Personalized Model (RFM + Apriori) Performance:

Metric	Value
Validation Method	Leave-Last-Out
Effective Validation Samples	1,296 customers
Hits (≥1 match in Top-5 recommendations)	264 orders
Hit Rate @ K=5	**20.37%**

Table 14

Model Validation Metrics

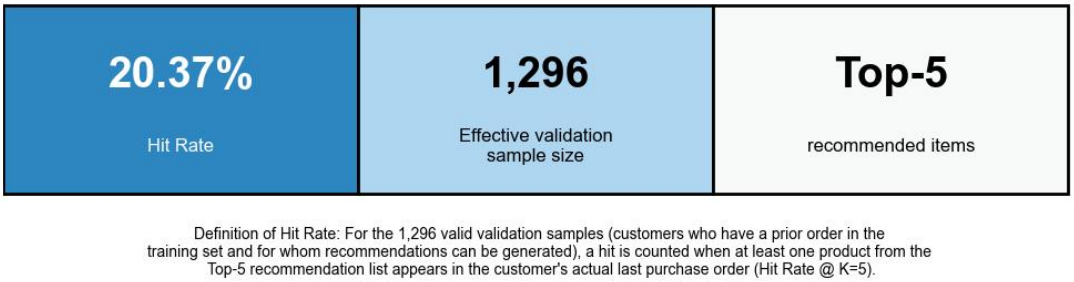


Figure:6

Definition of Hit Rate: For the 1,296 valid validation samples (customers who have a prior order in the training set and for whom recommendations can be generated), a hit is counted when at least one product from the Top-5 recommendation list appears in the customer's actual last purchase order.

8.2.4 Recommendation Coverage Statistics

Overall Coverage:

Metric	Value
Total Active Customers	4,335
Customers with Successful Recommendations	1,830
Overall Recommendation Coverage	**42.2%**
Number of Recommended Items per Customer	Top-3

Table 15

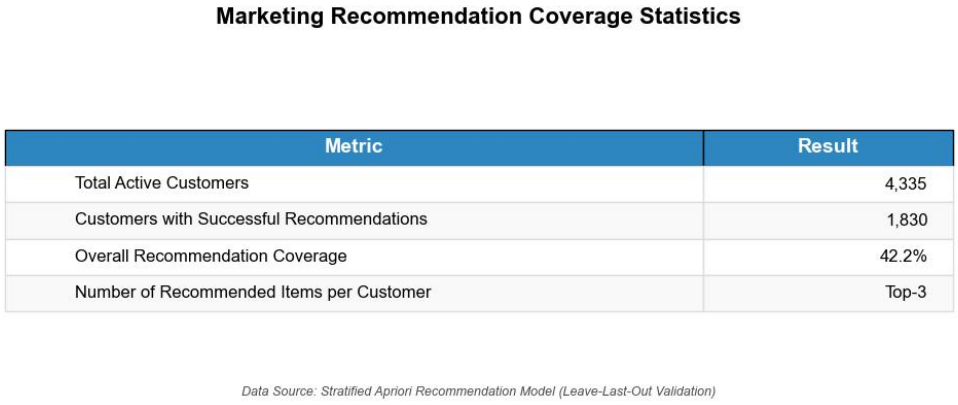


Figure:7

Coverage by Segment:

Segment	Total Customers	Customers with Recommendations	Coverage Rate
VIP	944	523	55.4%
Consistent Buyers	454	256	56.4%
Inactive	2,026	810	40.0%
Potential Loyalists	911	241	26.5%
Total	**4,335**	**1,830**	**42.2%**

Table 16

Segment Recommendation Coverage Rates

segment	recommendation coverage rates
VIP	**55.4%**
Consistent Buyers	56.4%
Inactive	40.0%
Potential Loyalists	26.5%

Data Source: Stratified Apriori Model - Marketing Recommendations Output

Figure:8

Segment Recommendation Coverage Statistics

Segment	Total Customers	Customers with Recommendatio	Coverage Rate
Consistent Buyers	454	256	56.4%
VIP	944	523	55.4%
Inactive	2,026	810	40.0%
Potential Loyalists	911	241	26.5%
Total	**4,335**	**1,830**	**42.2%**

Data Source: generate_marketing_list function execution log

Figure:11

Key Observation: High-value segments (VIP, Consistent Buyers) achieve >55% coverage, while Potential Loyalists have only 26.5% coverage due to sparse/dispersed purchase histories.

8.2.5 Comparative Analysis: Personalized Model vs. Global Baseline

Baseline Strategy: Recommend the 5 most frequently sold items across the entire dataset to every customer, regardless of segment or purchase history.

Global Top-5 Items: 85123A, 22423, 85099B, 84879, 47566

Metric	**Personalized (RFM + Apriori)**	**Global Baseline (Top-5)**
Hit Rate @ K=5	20.37%	16.09%
Effective Validation Samples	1,296	3,032
Hit Orders	264	488
Relative Improvement	**+26.6%**	–

Table 17

Relative Improvement Calculation: $$\text{Relative Improvement} = \frac{20.37% - 16.09%}{16.09%} \times 100% = 26.6%$$

Strategic Implication: While the Global Top-5 strategy reaches more customers (3,032 vs. 1,296), the personalized approach achieves significantly higher precision for customers where recommendations can be generated.

8.2.6 Sample Recommendation Cases (from Validation)

CustomerID	Segment	Input Basket (Previous Purchase)	Recommended Items	Hit Items
12793	Inactive	[21992, 22383, 20728]	[20725]	[20725] ✓
14670	Inactive	[21922, 22967, 22804]	[20675, 20676, 22522, 22578]	[20675, 20676] ✓
14465	Inactive	[22386, 85099B, 21931]	[22634, 82581, 84879, 85123A]	[22634, 85123A] ✓
17991	Inactive	[22697, 22698]	[22423, 22699]	[22423] ✓
14304	Inactive	[85049G, 22311, 21591]	[22138]	[22138] ✓

Table 18

8.3 Exploratory Data Analysis – Additional Findings

8.3.1 Seasonal Revenue Pattern

Peak Season: September – November (holiday shopping period)

Period	Estimated Revenue (£)	% of Annual Total
Sep – Nov (Peak)	~3,070,000	~35%
Dec – Aug (Regular)	~5,700,000	~65%
Annual Total	**~8,770,000**	**100%**

Table 19

Recommendation: Future iterations should create "Holiday Rule Sets" with time-decay weighted rules to capture seasonal buying patterns (e.g., gift items, party supplies, decorations).

8.3.2 Geographic Revenue Distribution

Country	Revenue Contribution
United Kingdom	~85% (>£7.3M)
Netherlands	~3%
Germany	~2%
France	~2%
Others (51 countries)	~8%

Table 20

Implication: Marketing efforts should prioritize UK market; international expansion requires market-specific strategies.

8.3.3 Top 10 Products by Revenue

Rank	Product Description	StockCode
1	REGENCY CAKESTAND 3 TIER	22423
2	WHITE HANGING HEART T-LIGHT HOLDER	21094
3	JUMBO BAG RED RETROSPOT	20725
4	PARTY BUNTING	21080
5	ASSORTED COLOUR BIRD ORNAMENT	20914
6	PAPER CRAFT LITTLE BIRDIE	22492
7	MEDIUM CERAMIC TOP STORAGE JAR	21086
8	RABBIT NIGHT LIGHT	21913
9	PAPER CHAIN 50'S CHRISTMAS	22138
10	NATIVITY SCENE	22424

Table 21

Observation: Top products are predominantly home decor and party supplies, indicating product category concentration risk.

8.4 Timeline Summary

Phase	Activity	Duration	Deadline
1. Proposal	Finalize topic, define business objectives, submit Project Proposal	7 Days	Mar 06
2. Data Preprocessing	Data cleaning, handling missing values, data type formatting	5 Days	
3. EDA & Visualization	Descriptive analysis, distribution checks, key metric visualization	4 Days	
4. Analysis I (RFM)	RFM score calculation, customer segmentation, cluster profiling	7 Days	
5. Analysis II (Market Basket)	Apriori algorithm execution, rule generation, Hit & Cover rate calculation	7 Days	
6. Synthesis & Strategy	Business interpretation, strategic recommendations, draft Data Analysis report	9 Days	Apr 03
7. Refinement & Optimization	Analysis method optimization, visualization enhancement, detailed content adjustments	4 Days	
8. Presentation Preparation	PPT design, narrative structuring, presentation rehearsal	3 Days	Apr 23
9. Finalization & Submission	Supplementary analysis, formatting compliance, official report submission	7 Days	May 15

8.5 Figures List

Figure 1: Transaction-Level Descriptive Statistics

Clean

指标	count
Quantity	396507.0
UnitPrice	396507.0
lineamount	396507.0

Full

指标	count
Quantity	397924.0
UnitPrice	397924.0
lineamount	397924.0

指标	count	mean	std	min	25%	50%	75%	max
Quantity	396507.0	20.82	290.52	-80995.0	2.0	5.0	12.0	80995.0
UnitPrice	396507.0	3.82	24.34	0.0	1.25	1.95	3.75	3897.0
lineamount	396507.0	15.96	161.53	-8715.5	3.75	9.95	23.35	16857.0

Figure:1

Figure 2: Customer-Level Descriptive Statistics

Input Variables	count	mean	std	min	25%	50%	75%	max
Recency	4335.0	93.21	100.17	1.0	18.0	51.0	143.0	374.0
Frequency	4335.0	4.25	7.64	1.0	1.0	2.0	5.0	206.0
Monetary	4335.0	2022.55	8923.22	0.0	305.77	668.11	1631.48	279138.02

Figure:2

Figure 3: RFM Segmentation Rules & Business Meaning

Segment	Condition	Business Meaning
VIP	R ≥ 4 AND F ≥ 4 AND M ≥ 4	Recent, frequent, high-value core customers
Consistent Buyers	R ≥ 3 AND F ≥ 4 AND M ≥ 3	Recent, frequent loyal buyers
Potential Loyalists	R ≥ 2 AND F ≥ 3 AND M ≥ 2	Some purchase behavior, growth potential
Inactive	Otherwise	Low-frequency, low-spending, or churned customers

Figure:3

Figure 4: RFM Segmentation Results (Count & Revenue Share)

segment	customer_count	monetary_total	customer_pct	monetary_pct
VIP	944	5680666.020	21.77%	64.79%
Consistent Buyer	454	877151.771	10.47%	10.00%
Potential Loyalists	911	1134822.840	21.02%	12.94%
Inactive	2026	1075112.022	46.73%	12.26%

Figure:4

Figure 5: Recency-Frequency Behavioral Heatmap

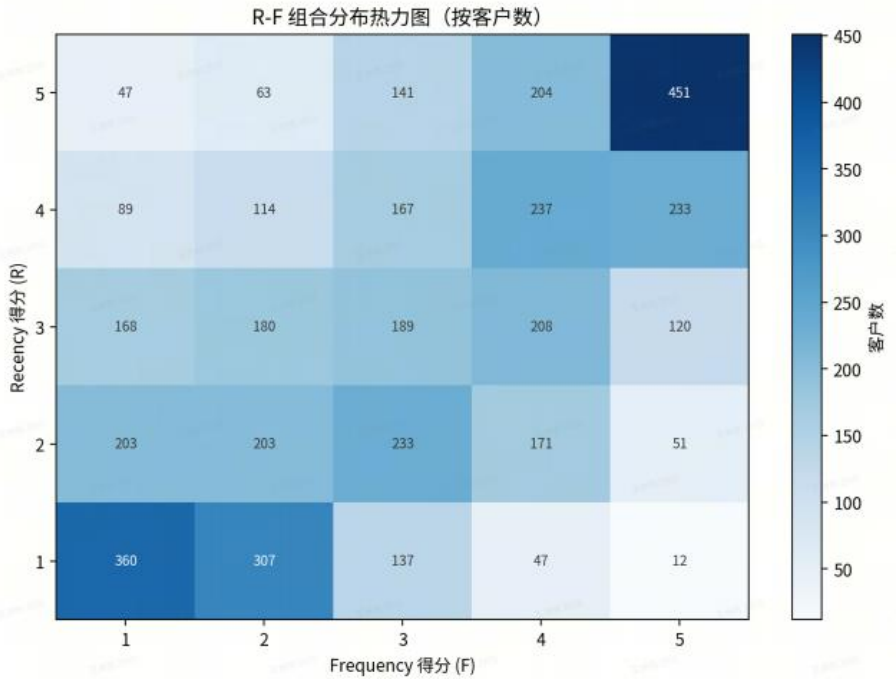
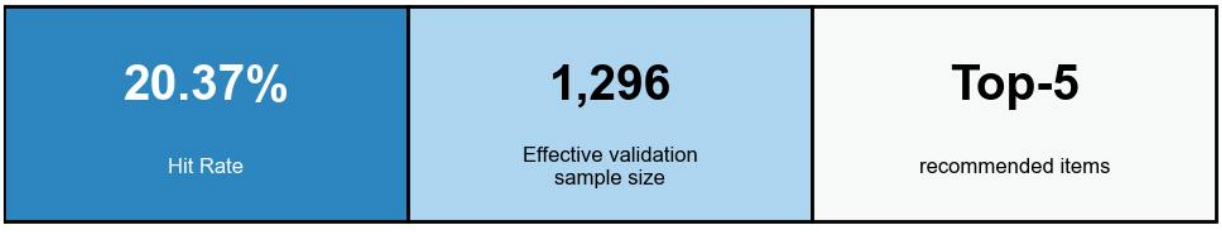


Figure:5

Figure 6: Model Validation Metrics



Definition of Hit Rate: For the 1,296 valid validation samples (customers who have a prior order in the training set and for whom recommendations can be generated), a hit is counted when at least one product from the Top-5 recommendation list appears in the customer's actual last purchase order (Hit Rate @ K=5).

Figure:6

Figure 7: Overall Recommendation Coverage Statistics

Marketing Recommendation Coverage Statistics

Metric	Result
Total Active Customers	4,335
Customers with Successful Recommendations	1,830
Overall Recommendation Coverage	42.2%
Number of Recommended Items per Customer	Top-3

Data Source: Stratified Apriori Recommendation Model (Leave-Last-Out Validation)

Figure:7

Figure 8: Recommendation Coverage Rates by Segment

Segment Recommendation Coverage Rates

segment	recommendation coverage rates
VIP	**55.4%**
Consistent Buyers	56.4%
Inactive	40.0%
Potential Loyalists	26.5%

Data Source: Stratified Apriori Model - Marketing Recommendations Output

Figure:8

Figure 9: Stratified Apriori Mining Strategy & Threshold Settings

Stratified Apriori Mining Strategy & Threshold Settings

Tier	Rule Base	Threshold Settings	Business Meaning
Tier 1	Segment-specific rule bases	VIP & Consistent Buyers: min_support=0.015, min_conf=0.18 Inactive & Potential Loyalists: min_support=0.005, min_conf=0.1	Capture differentiated purchasing habits: • High-Value: Stricter rules for precision • Low-Value: Looser rules for coverage
Tier 2 (Fallback)	Global rule base	min_support=0.01, min_confidence=0.15	Automatic downgrade when segment rules are empty, preventing gaps

Figure:9

Figure 10: Mining Results by Segment

Mining Results by Segment

Segment	Order Count	Min Support	Min Confidence	Rule Count
VIP	12,557	0.015	0.18	301
Consistent Buyers	2,795	0.015	0.18	165
Potential Loyalists	3,207	0.005	0.10	3,444
Inactive	3,217	0.005	0.10	1,835
Global (fallback)	-	0.01	0.15	632

Data Source: mine_segment_rules function execution log

Figure:10

Figure 11: Segment Recommendation Coverage Statistics

Segment Recommendation Coverage Statistics

Segment	Total Customers	Customers with Recommendatio	Coverage Rate
Consistent Buyers	454	256	56.4%
VIP	944	523	55.4%
Inactive	2,026	810	40.0%
Potential Loyalists	911	241	26.5%
Total	**4,335**	**1,830**	**42.2%**

Data Source: generate_marketing_list function execution log

Figure:11

8.6 Tables List

Table 1: Model Performance Comparison

Metric	Personalized (RFM + Apriori)	Global Baseline (Top-5)
Hit Rate	**20.37%**	16.09%
Valid Samples	1,296	3,032
Hit Orders	264	488

Table 1

Table 2: Data Source and Collection

Item	Description
Dataset Name	TATA Online Retail Data Set
Data Provider	Kaggle (Ishan Shrivastava)
Time Period	December 1, 2010 – December 9, 2011
Country of Origin	United Kingdom (dominant market)
Business Type	Non-store online retail, unique all-occasion gifts

Table 2

Table 3: Original Dataset Statistics

Metric	Value
Total Transaction Records	397,924
Unique Customer IDs (original)	4,338
Unique Products (StockCodes)	4,000+
Countries Represented	55

Table 3

Table 4: Data Cleaning Outcome

Metric	Value
Cleaned Transaction Records	396,507
Valid Unique Customers	4,335
Records Removed	1,417 (returns, fees, missing IDs)
Return Rate	2.2%

Table 4

Table 5: Data Dictionary

Variable Name	Data Type	Description	Valid Range / Notes
CustomerID	Integer	Unique identifier for each customer	12346 – 18287
InvoiceNo	String	Transaction session identifier	6-digit integer; C-prefix = cancellation
InvoiceDate	Datetime	Transaction timestamp	2010-12-01 to 2011-12-09
StockCode	String	Product identifier	5-digit alphanumeric code
Description	String	Product name/description	4,000+ unique items
Quantity	Integer	Number of units per transaction line	-80,995 to 80,995 (negative = return)
UnitPrice	Float	Price per unit in British Pounds (GBP)	£0.00 to £649.50
LineAmount	Float (Derived)	Quantity × UnitPrice (revenue per line)	£0.00 to £168,469.60
Country	String	Customer geographic location	55 countries; UK dominant (~85% revenue)

Table 5

Table 6: Data Cleaning Procedure

Step	Action	Records Affected	Rationale

Step	Action	Records Affected	Rationale
1	Remove missing CustomerID	135,239 records	Cannot analyze anonymous transactions for customer-level analytics
2	Remove missing Description	1,454 records	Cannot identify products for recommendation
3	Remove returns (Quantity < 0)	8,966 records	Not genuine sales; would distort Monetary values
4	Remove cancellations (InvoiceNo starts with "C")	7,745 records	Transaction voided; not valid purchases
5	Remove non-product fees	3,013 records	POSTAGE, BANK, DISCOUNT, AMAZON FEE, etc. are not product revenue
6	Remove zero/negative UnitPrice	234 records	Data errors or promotional items

Table 6

Table 7: Transaction-Level Statistics (Cleaned Data, N = 396,507)

Variable	Count	Mean	Std. Dev.	Min	25%	50% (Median)	75%	Max
Quantity	396,507	13.04	180.70	1	2	6	12	80,995
UnitPrice (£)	396,507	2.88	4.35	0.00	1.25	1.95	3.75	649.50
LineAmount (£)	396,507	22.11	308.83	0.00	4.68	11.80	19.80	168,469.60

Table 7

Table 8: Customer-Level Statistics (RFM Variables, N = 4,335)

Variable	Count	Mean	Std. Dev.	Min	25%	50% (Median)	75%	Max
Recency (days)	4,335	93.21	100.17	1	18	51	143	374
Frequency	4,335	4.25	7.64	1	1	2	5	206
Monetary (£)	4,335	2,022.55	8,923.22	0	305.77	668.11	1,631.48	279,138.02

Table 8

Table 9: RFM Scoring Methodology

Component	Scoring Method	Scale	Interpretation
R (Recency)	Reverse quintile scoring	1–5	5 = most recent purchase; 1 = longest ago
F (Frequency)	Quintile scoring	1–5	5 = highest frequency; 1 = lowest frequency
M (Monetary)	Quintile scoring	1–5	5 = highest spend; 1 = lowest spend
Total Score	R + F + M	3–15	Higher score = higher customer value

Table 9

Table 10: RFM Score Distribution

Percentile	RFM Total Score Threshold
25th	≤ 7
50th	≤ 9
75th	≤ 12
90th	≥ 13

Table 10

Table 11: RFM Segmentation Results

Segment	Customer Count	Customer %	Revenue (£)	Revenue %	Avg Revenue/Customer (£)
VIP	944	21.77%	5,680,666.02	64.79%	6,017.44
Consistent Buyers	454	10.47%	877,151.77	10.00%	1,932.05
Potential Loyalists	911	21.02%	1,134,822.84	12.94%	1,245.69
Inactive	2,026	46.73%	1,075,112.02	12.26%	530.66
Total	**4,335**	**100.00%**	**8,767,752.65**	**100.00%**	**2,022.55**

Table 11

Table 12: Stratified Apriori Mining Strategy

Segment Group	Min Support	Min Confidence	Business Logic
VIP & Consistent Buyers	0.015	0.18	Stricter rules for precision; ensure high relevance for premium users
Inactive & Potential Loyalists	0.005	0.10	Looser rules for coverage; capture weak patterns in sparse data
Global (Fallback)	0.01	0.15	Safety net for cold-start scenarios when segment rules are empty

Table 12

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Consistent Buyers	2,795	0.015	0.18	165
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Global (Fallback)	–	0.01	0.15	632

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Table 14: Personalized Model (RFM + Apriori) Performance

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Table 14

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Inactive	2,026	810	40.0%
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Total	**4,335**	**1,830**	**42.2%**

Table 16

Table 17: Comparative Analysis (Personalized Model vs. Global Baseline)

Metric	Personalized (RFM + Apriori)	Global Baseline (Top-5)
Hit Rate @ K=5	20.37%	16.09%
Effective Validation Samples	1,296	3,032
Hit Orders	264	488
Relative Improvement	**+26.6%**	–

Table 17

Table 18: Sample Recommendation Cases (from Validation)

CustomerID	Segment	Input Basket (Previous Purchase)	Recommended Items	Hit Items
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14465	Inactive	[22386, 85099B, 21931]	[22634, 82581, 84879, 85123A]	[22634, 85123A] ✓
17991	Inactive	[22697, 22698]	[22423, 22699]	[22423] ✓
14304	Inactive	[85049G, 22311, 21591]	[22138]	[22138] ✓

Table 18

Table 19: Seasonal Revenue Pattern

Period	Estimated Revenue (£)	% of Annual Total
Sep – Nov (Peak)	~3,070,000	~35%
Dec – Aug (Regular)	~5,700,000	~65%
Annual Total	**~8,770,000**	**100%**

Table 19

Table 20: Geographic Revenue Distribution

Country	Revenue Contribution
United Kingdom	~85% (>£7.3M)
Netherlands	~3%
Germany	~2%
France	~2%
Others (51 countries)	~8%

Table 20

Table 21: Top 10 Products by Revenue

Rank	Product Description	StockCode
1	REGENCY CAKESTAND 3 TIER	22423
2	WHITE HANGING HEART T-LIGHT HOLDER	21094

Rank	Product Description	StockCode
3	JUMBO BAG RED RETROSPOT	20725
4	PARTY BUNTING	21080
5	ASSORTED COLOUR BIRD ORNAMENT	20914
6	PAPER CRAFT LITTLE BIRDIE	22492
7	MEDIUM CERAMIC TOP STORAGE JAR	21086
8	RABBIT NIGHT LIGHT	21913
9	PAPER CHAIN 50'S CHRISTMAS	22138
10	NATIVITY SCENE	22424

Table 21